

Shri G. S. Institute of Technology & Science, Indore

Department of Mechanical Engineering

Date: 20/12/2021

Mid-Term Test-I

Session (2021-2022) Semester-1

Subject: Fundamental of Mechanical Engineering (ME-10650)

Class: I Year

Max Marks: 20

Time: 1 hour

Note: Attempt any four questions, all questions carry equal marks.

		Marks	CO	BL	PI
Q. No. 1	Explain the First Law of Thermodynamics as referred to closed systems undergoing a cyclic change.	5	1	L2	1.3.1
Q. No. 2	A stream of gases at 7.5 bar, 750°C and 140 m/s is passed through a turbine of a jet engine. The stream comes out of the turbine at 2.0 bar, 550°C and 280 m/s. The process may be assumed adiabatic. The enthalpies of gas at the entry and exit of the turbine are 950 kJ/kg and 650 kJ/kg of gas respectively. Determine the capacity of the turbine if the gas flow is 5 kg/s.	5	1	L3	1.4.1
Q. No. 3	Define Newtonian and Non-Newtonian fluids.	5	2	L2	1.3.1
Q. No. 4	Water is flowing through a pipe having diameter 300 mm and 200 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 24.525 N/cm ² and the pressure at the upper end is 9.81 N/cm ² . Determine the difference in datum head if the rate of flow through pipe is 40 lit/s.	5	2	L3	1.4.1
Q. No. 5	Discuss in brief with their function (a) Fusible plug (b) Blow of cock (c) Feed check valve	5	5	L2	1.3.1
Q. No. 6	Explain the working of Babcock and Wilcox boiler with the help of a neat sketch.	5	5	L2	1.3.1

CO: Course Outcomes, BL: Bloom's taxonomy Level, PI: Performance Indicators